Continually Learning Agents

Seminar | Saarland University | Winter Semester 2026/27

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Course information

Credits	4 CP (or 7 CP)
Lecture	Wednesday, 12:15-13:45
Location	Room 1.12, C7.3
Website	https://github.com/at-lab-uds/continually-learning-agents-seminar
CMS	https://cms.sic.saarland/cla/
Forum	https://cms.sic.saarland/cla/forum/
LSF	Here

Target audience

- Master students and advanced Bachelor students
- Some background in ML and NLP
- Good understanding of LMs
- Limited spots available, assignment will be made later this week! ⚠️

Format and grading

- Each student gives one presentation, we all read the papers and discuss
- ⚠️ AI generated slides are not accepted! Create your own slides!
  - Making slides is important for shaping your understanding of the work
  - I want to hear your thoughts, not Claude's or GPT's
- We will have 1-2 presentations per week, depending on the paper

Grading

- 4 CP version
  - Presentation: 70%
  - Participation in class: 30%
- 7 CP version
  - Presentation: 45%
  - Participation in class: 15%
  - Final paper: 40%

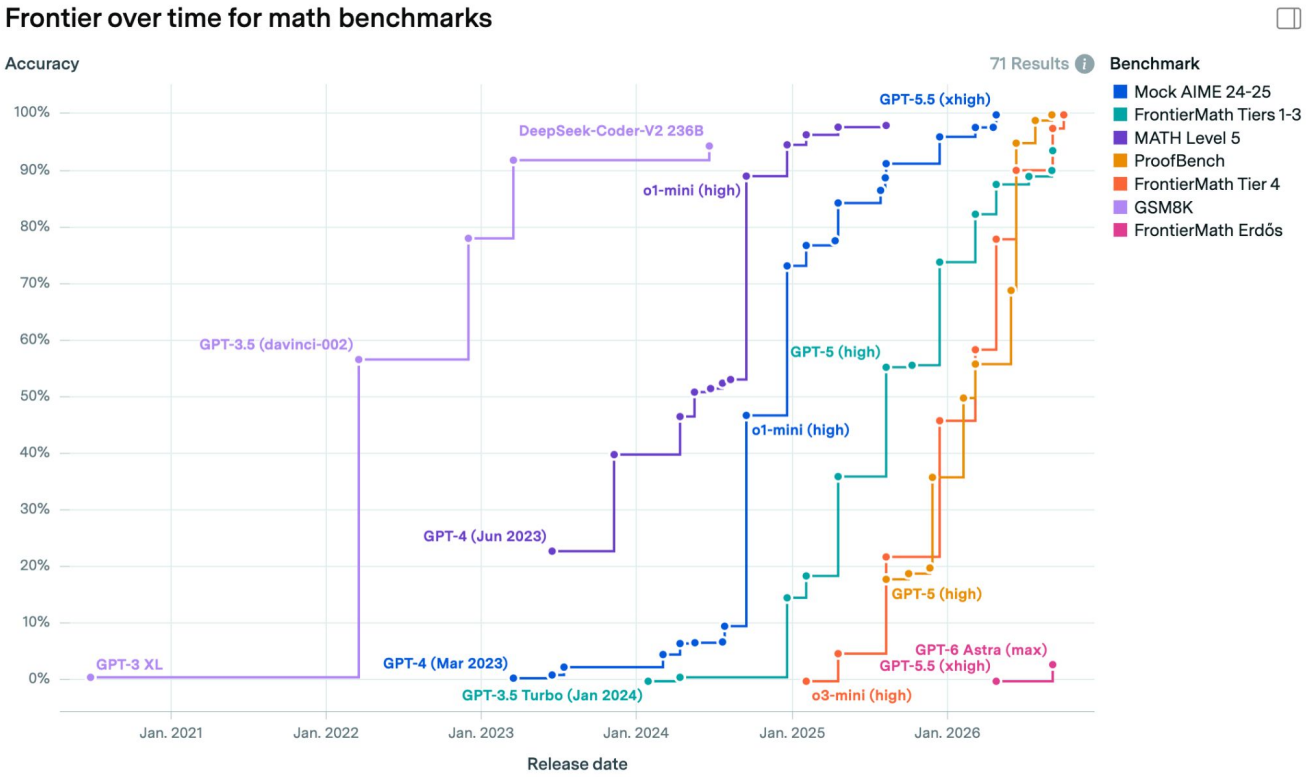
Course schedule

#	Date	Topic	Papers	Presenters
1	14 Oct 2026	Kick-off meeting: introduction and course logistics	—	—
2	21 Oct 2026	No class	—	—
3	28 Oct 2026	*TBA*	*TBA*	*TBA*
4	04 Nov 2026	*TBA*	*TBA*	*TBA*
5	11 Nov 2026	*TBA*	*TBA*	*TBA*
6	18 Nov 2026	*TBA*	*TBA*	*TBA*
7	25 Nov 2026	*TBA*	*TBA*	*TBA*
8	02 Dec 2026	*TBA*	*TBA*	*TBA*
9	09 Dec 2026	*TBA*	*TBA*	*TBA*
10	16 Dec 2026	*TBA*	*TBA*	*TBA*
11	06 Jan 2027	*TBA*	*TBA*	*TBA*
12	13 Jan 2027	*TBA*	*TBA*	*TBA*
13	20 Jan 2027	*TBA*	*TBA*	*TBA*
14	27 Jan 2027	*TBA*	*TBA*	*TBA*
15	03 Feb 2027	*TBA*	*TBA*	*TBA*

- Wednesdays, 12:15 - 13:45
- **No class next week!** ⚠️
- Schedule is tentative
- Changes will be communicated via the forum and github page
- Paper deadline: TBD

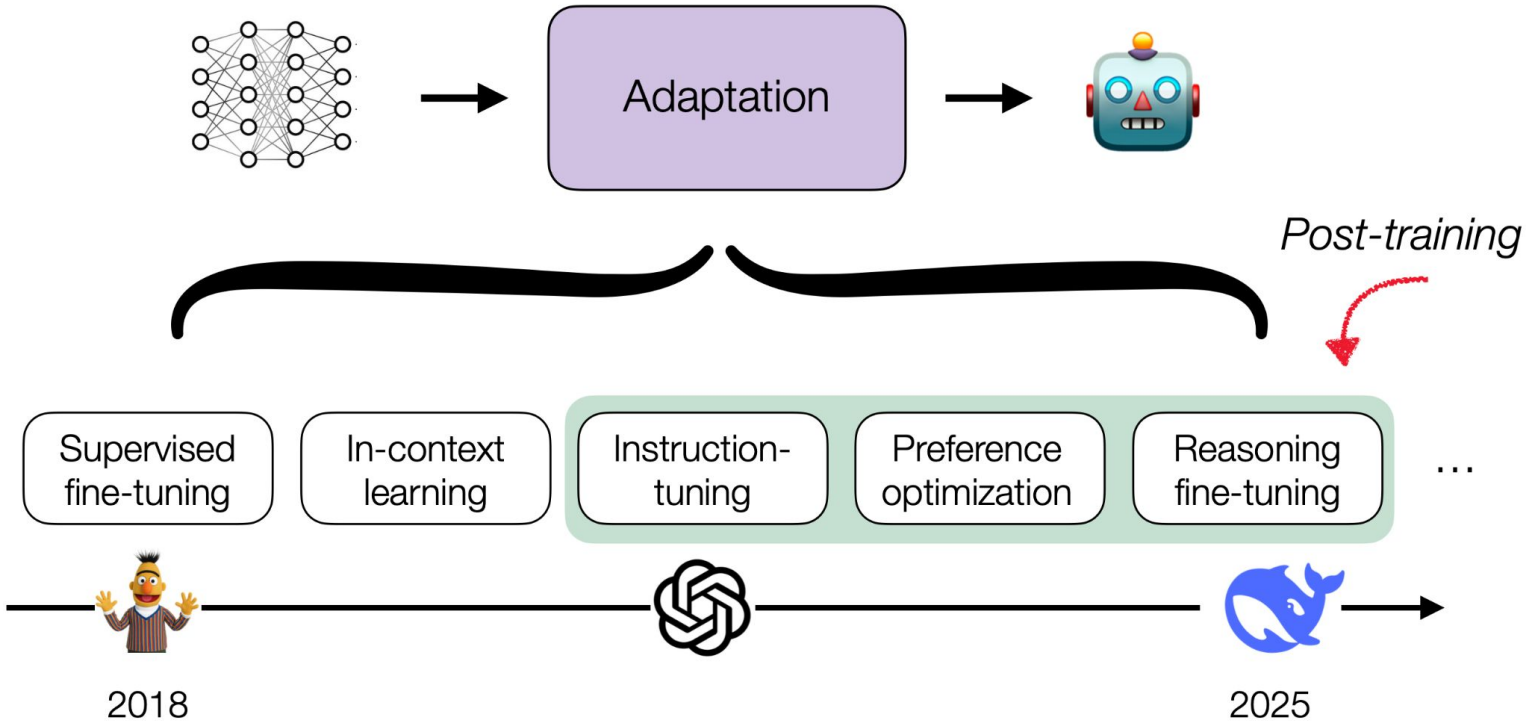
Why continual learning?

AI progress

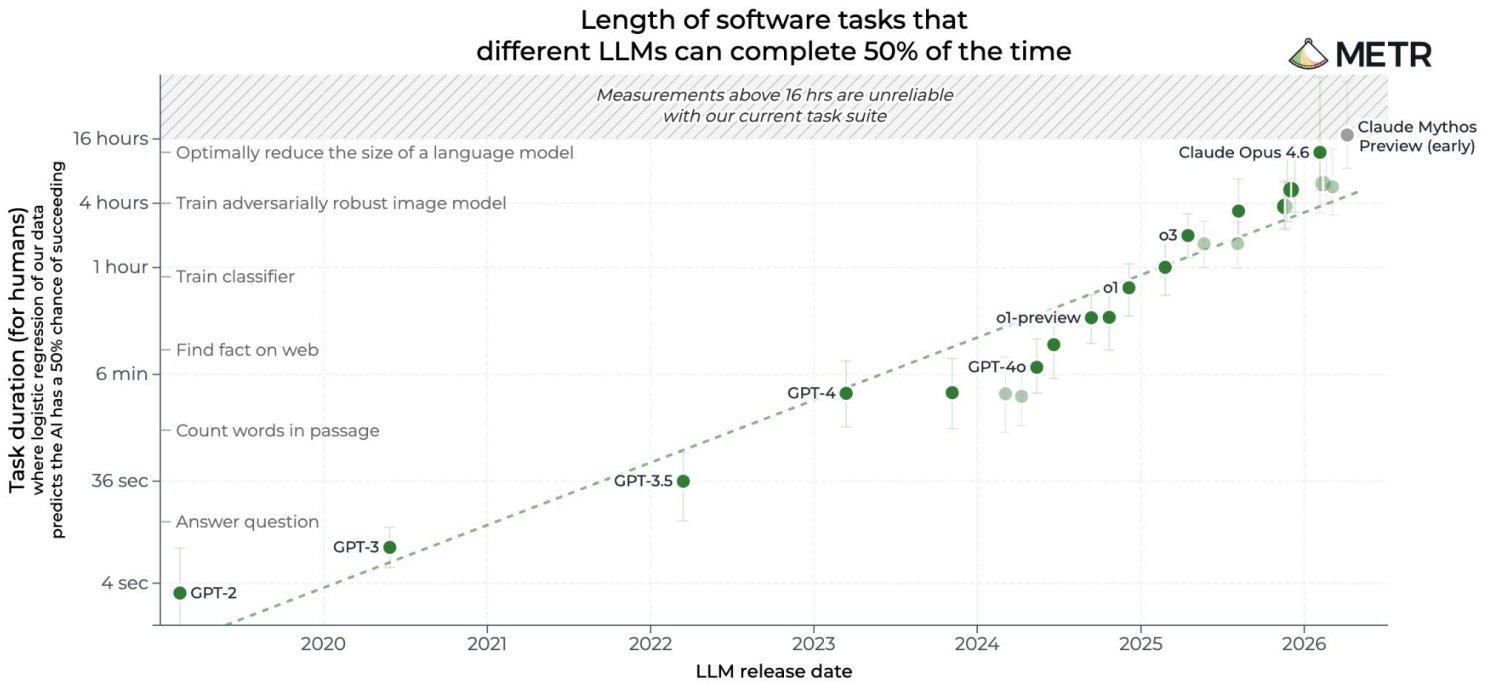


Source: https://epoch.ai/benchmarks?view=graph&tab=eci

How are these models trained?



Long horizon use cases



Source: https://metr.org/time-horizons/

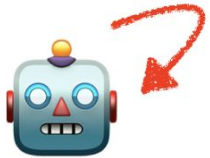
How can we persist these new *experiences*?

- Broadly speaking, two classes of approaches
  - Weight-based learning
  - Context-based learning

What are the challenges with these?

- *Catastrophic forgetting*
- Limited context
- Changes are too dramatic

The promise of continual learning 🔮



Future paradigm: continually updating AI systems learning from “experience”

Prediction: We will see models that …

Are truly **personalized** systems	Effectively acquire, edit, and generate (new) **knowledge**	Autonomously perform **long horizon** tasks

Thank you!

Next lecture: 28.10.2026

Important next steps for you: Sign up in the CMS; Send me your paper preferences

Questions? 👉 Use the forum: https://cms.sic.saarland/cla/forum/ if you think it's relevant for others too. Otherwise, send me an email: marius.mosbach@uni-saarland.de (use [Continually Learning Agents] in subject line)